

**Machine-Learning & DFT-Guided Electrode
Design for Oxide-based Supercapacitors: From
Data Mining to Experimental Validation**

Background

The Problem

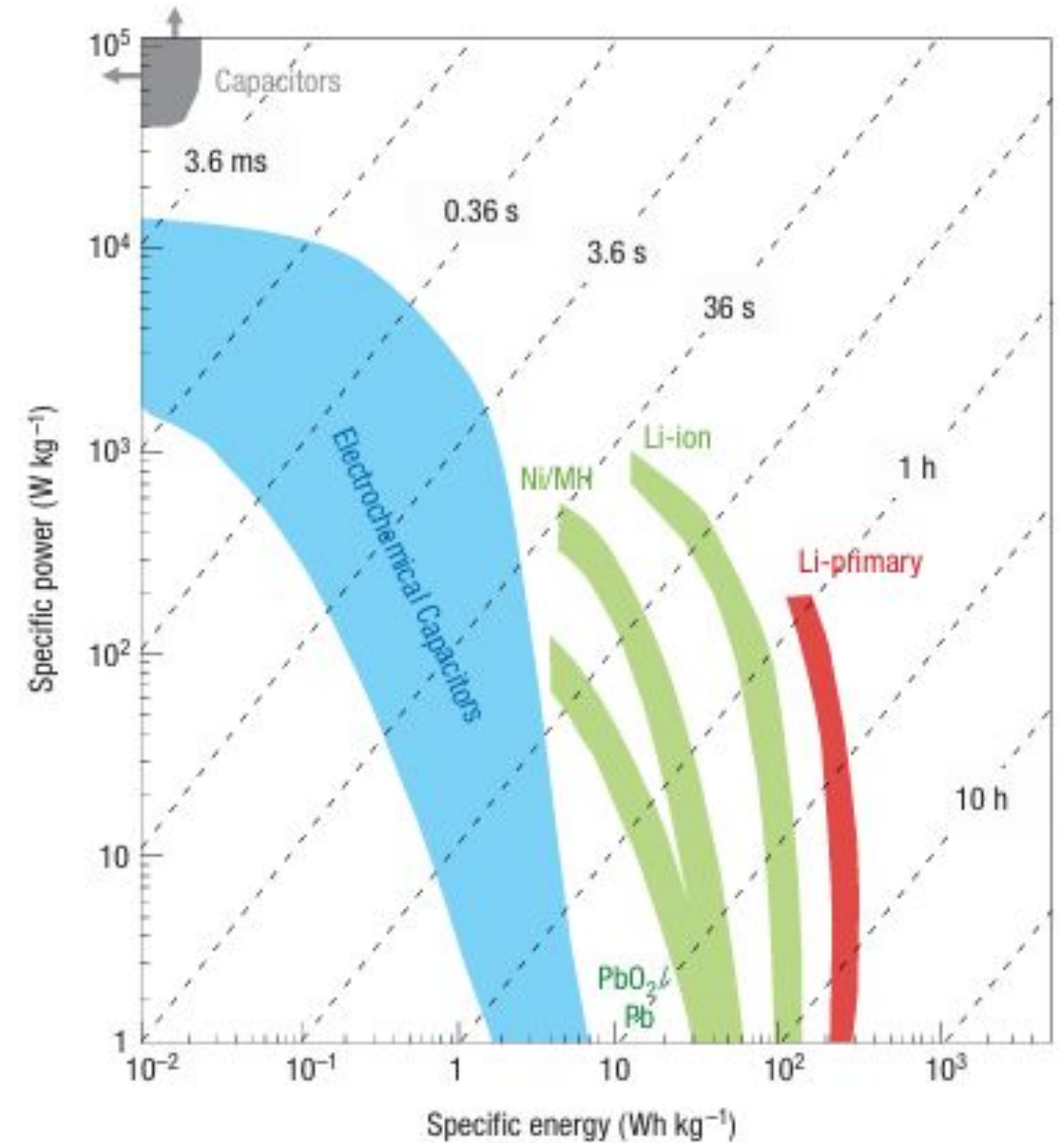
Capacitors: high power density, low energy density.

Batteries: high energy density, low power density.

The Solution

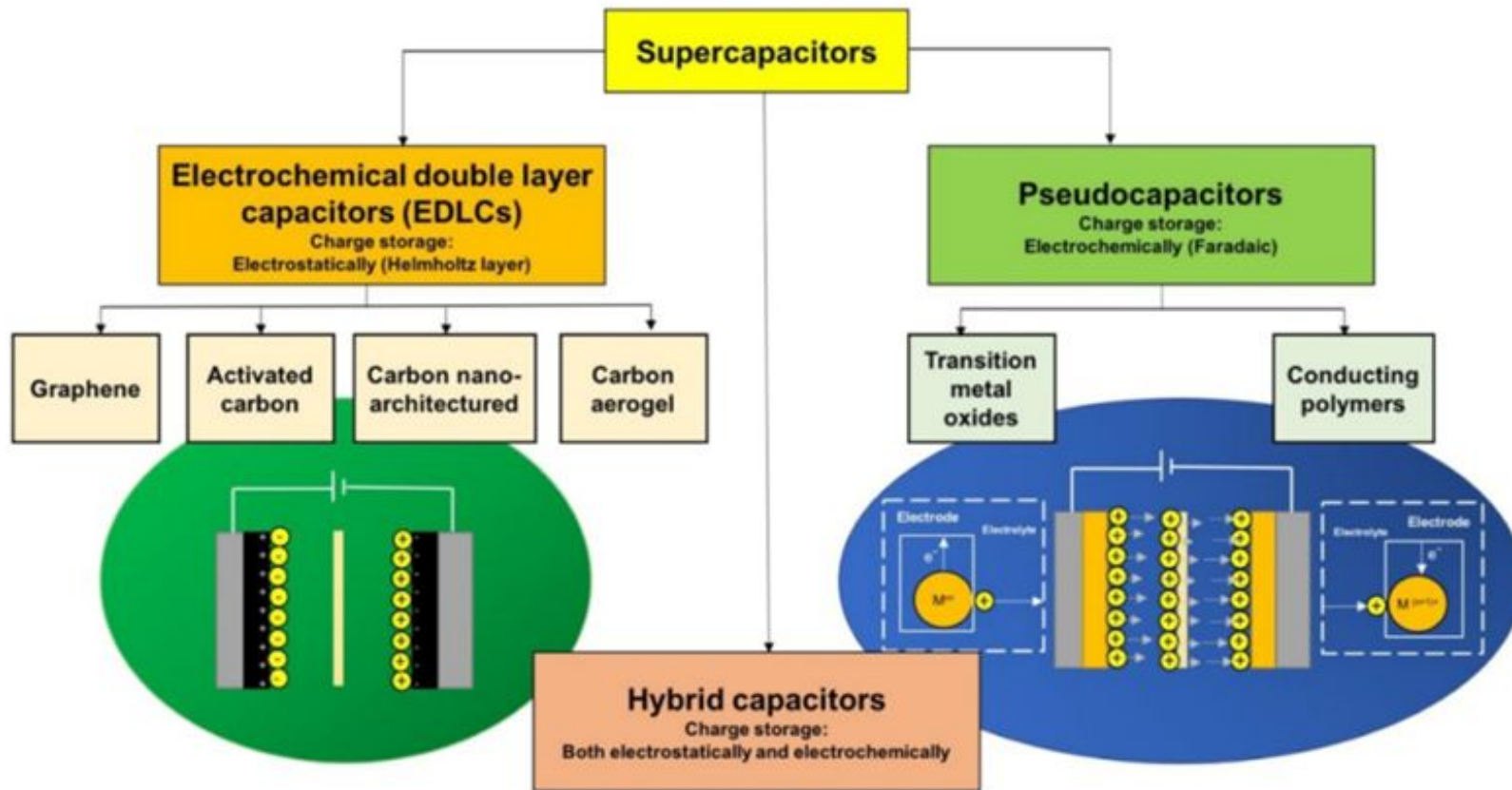
Supercapacitors can achieve —

- Best of both between power & energy density
- Faster charge/discharge cycle
- Longer cycle life



Specific power vs. specific energy plot for energy-storage devices (Ragone plot)

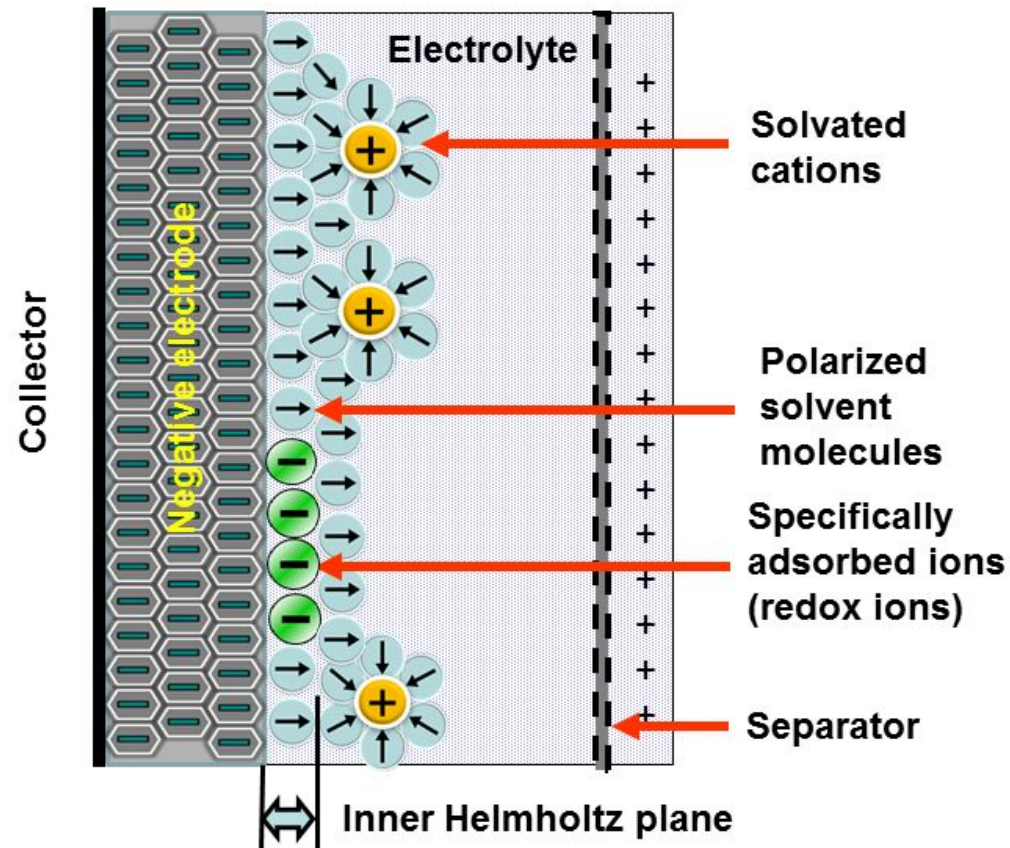
Supercapacitors



Charge-storage Mechanisms	Description
EDLC	Electrostatic storage; high-surface-area carbon electrodes
Pseudo-capacitance	Reversible Faradaic redox; TMOs & conducting polymers
Hybrid	Combines both

Transition Metal Oxides: Material-type of Choice

Pseudocapacitance with specifically adsorbed ions



→ Excellent redox behaviour

→ High **pseudocapacitive** performance

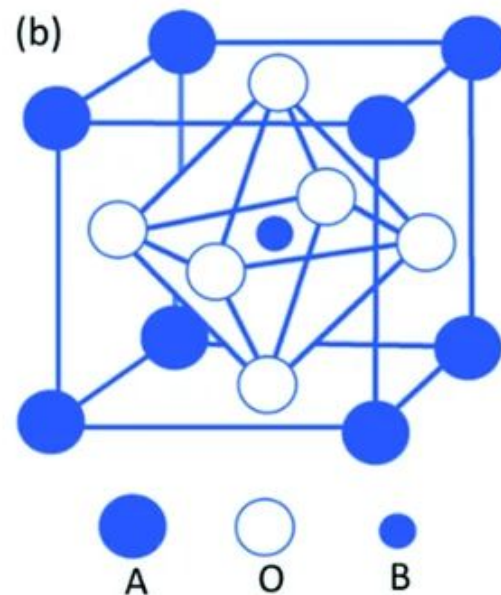
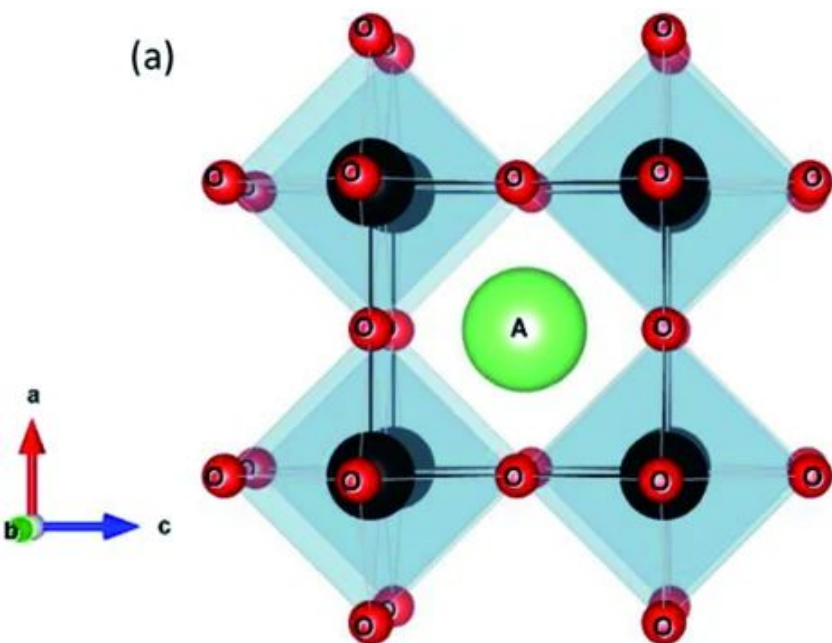
→ High specific capacitance

→ Combined effects from multiple transition metals

→ Better performance in energy density, power density, and cycle stability

→ Several Available structures: Spinel (AB_2O_4), Rock Salt (AO), Perovskites (ABO_3 , $ABB'O_6$) etc.

Perovskite Oxides: Structure of Choice



- Structural flexibility
- Stable structure
- Better cycling stability
- Mixed ionic-electronic conductivity
- Multi-electron transfer during redox reactions
- large specific pseudocapacitance

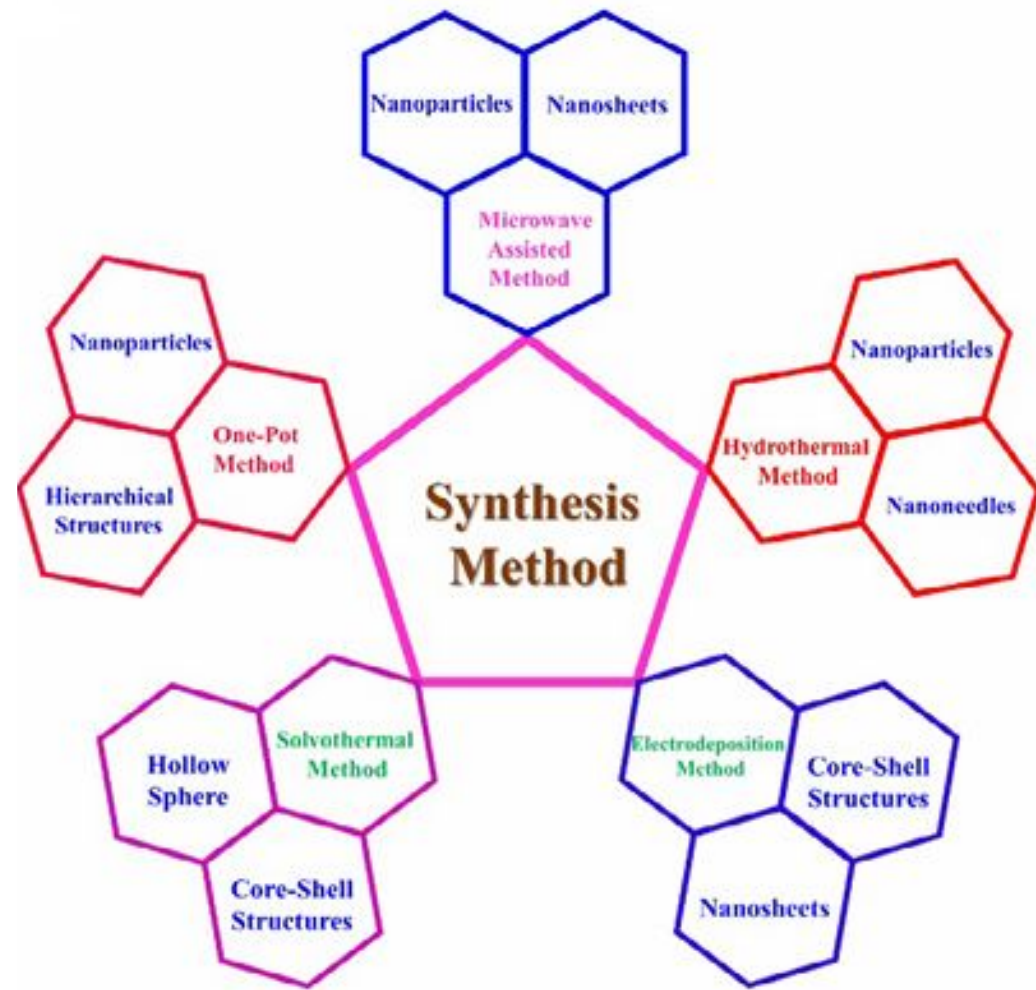
Perovskite Structure - ABO_3

- A-site: larger cation (e.g., La, Sr, Bi)
- B-site: smaller transition metal cation (e.g., Mn, Co, Ni, Fe)
- O: oxygen ions

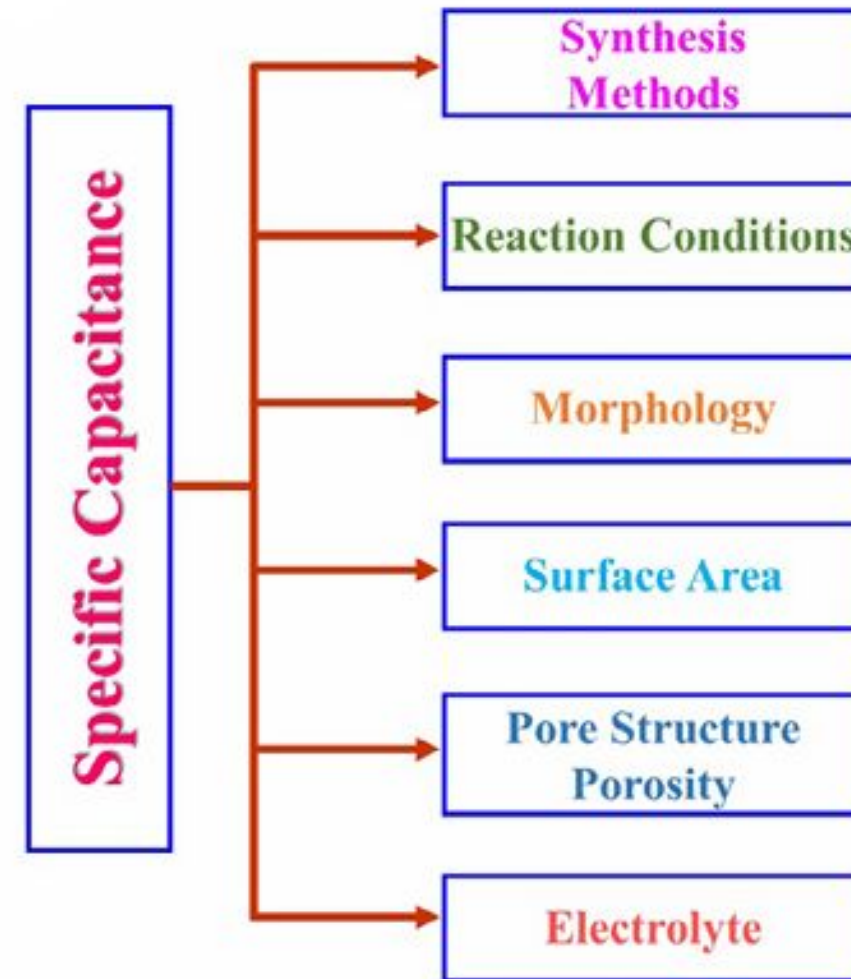
Electro-Design Optimization Challenge

Conventional Trial-and-error methods	Computation-Guided Closed-loop method
Slow	Fast
High Cost	Low Cost
Intuition-Based Guesswork	Analysis Based
No Feedback	Feedback Mechanism
Uncertain Outcome	Higher level of accuracy in outcome
Synthesis-first; understand-later approach	Predictive screening before synthesis

Electrode-Design Optimization Challenge



Synthesis method vs. observed morphology



Parameters influencing specific capacitance

Literature Review

Check for updates

ChemPhysChem

Research Article
 doi.org/10.1002/cphc.202400629

Chemistry Europe
 European Chemical Societies Publishing

www.chemphyschem.org

Machine Learning-Based Prediction of Supercapacitor Capacitance for MgCo_2O_4 Electrodes

Mengfan Tang,^[a] Yue Ding,^[a] Tanwei Hu,^[a] Xiaolong Zhu,^[a] Guang Zheng,^[a] and Yu Tian^{*[a]}

Electrode materials are essential in the electrochemical process of storing charge in supercapacitors and have a significant impact on the cost and capacitive performance of the final product. Hence, it is imperative to make precise predictions regarding the capacitance of electrode materials in order to further the development of supercapacitors. MgCo_2O_4 , with a theoretical capacitance of up to 3122 F g^{-1} , holds immense research value as an electrode material. The objective of this study is to predict the capacitance of MgCo_2O_4 with high accuracy. This will be achieved by extracting numerous data from published papers and using some parameters as input features. The Recursive Feature Elimination (RFE) method was employed, using Random Forest (RF), Extreme Gradient Boosting (XGBoost) and Regression Tree (RT) as selectors to identify the optimal feature subset. Then, combining them with these three regression models to construct nine machine learning (ML) models. After performance evaluation and outlier analysis, the XGB-RFE-XGB model achieved R-squared (R^2), root mean squared error (RMSE), and mean absolute error (MAE) of 0.95, 111.83 F g^{-1} and 68.25 F g^{-1} , respectively, demonstrating its stability and reliability. Therefore, the XGB-RFE-XGB model can be used as a reliable predictive tool in subsequent experimental designs.

Check for updates

Energy Advances

ROYAL SOCIETY OF CHEMISTRY

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PAPER

A machine learning approach for estimating supercapacitor performance of graphene oxide nano-ring based electrode materials†

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This work utilizes a novel approach leveraging the machine learning (ML) technique to predict the electrochemical supercapacitor performance of graphene oxide nano-rings (GONs) as electrode nanomaterials. Initially, the experimental procedure was carried out to synthesize GO via a modified Hummers method, followed by GONs preparation using the water-in-oil (W/O) emulsion technique. High-resolution transmission electron microscopy (HRTEM) analysis reveals the formation of a typical two-dimensional GO nanosheet and multilayer-GO nano-rings. The X-ray diffraction (XRD), Raman spectroscopy, X-ray photoelectron spectroscopy (XPS), and Brunauer–Emmett–Teller (BET) analysis results show that the GONs possess similar structural and surface chemistry properties as of GO, with a slight reduction in oxygenous functionalities, enhancing the capacitive behaviours through facile electron migration at the electrode surface. The electrochemical assessment of GO and GONs samples indicates outstanding specific capacitances of 164 F g^{-1} and 294 F g^{-1} at 1 mV s^{-1} , showcasing capacitive retention of up to 63% and 60% after 2500 cycles. In addition, four different machine learning models were tested to estimate the role of electrochemical parameters in determining the specific capacitance of GONs.

Cite this: *Energy Adv.*, 2025, 4, 119
 Received 23rd October 2024, Accepted 20th November 2024
 DOI: 10.1039/d4ya00577e
 rsc.li/energy-advances

Research Gap
 Limited to a single material class
 No Process level descriptors

Research Gap
 Restricted to a single morphology
 Applicable to single material system

Literature Review

Journal of Power Sources 665 (2026) 239040



Contents lists available at ScienceDirect

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour



Machine learning and explainable artificial intelligence reveal physical insights into biomass-derived carbon for high-performance supercapacitors

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HIGHLIGHTS

- ML models gave $R^2 = 0.92$ (capacitance) and $F1 = 0.73$ (cyclic stability).
- XAI used to reveal mechanisms behind biomass carbon supercapacitor performance.
- Flash pyrolysis boosts capacitance, balancing surface area and stability.

GRAPHICAL ABSTRACT



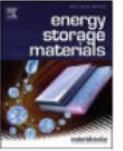
Energy Storage Materials 40 (2021) 426–438



Contents lists available at ScienceDirect

Energy Storage Materials

journal homepage: www.elsevier.com/locate/ensm



Machine learning-based prediction of supercapacitor performance for a novel electrode material: Cerium oxynitride

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ARTICLE INFO

Keywords:
Machine learning
Cerium
Oxynitride
Prediction
Electrode
Supercapacitor

ABSTRACT

From an engineering standpoint, specific capacity and cyclic stability may be considered the two most critical performance-related features for supercapacitor electrodes. The prediction of these two parameters is hence crucial for evaluating the prospect of a given material for a supercapacitor-electrode application. However, this prediction is highly non-trivial using existing atomistic approaches. As a solution, a combinatorial approach of value and grade prediction machine-learning models are used to predict the performance of a novel material (cerium oxynitride) for supercapacitor application. The model predicts the material to have a specific capacity of $\sim 26.6 \text{ mAh g}^{-1}$ and capacity retention of $>90\%$ for a particular material (morphology, composition, surface area) and operational (current density, applied potential window etc.) properties; which can be viably achieved via urea glass method. The experimental results ($\sim 26 \text{ mAh g}^{-1}$ and $\sim 100\%$ capacity retention) considerably validate the predictive approach presented here. This article is the first instance wherein cerium oxynitride has been predicted and reported as a supercapacitor electrode. This makes the prediction and the validation made in this study of contemporary relevance.

Research Gap

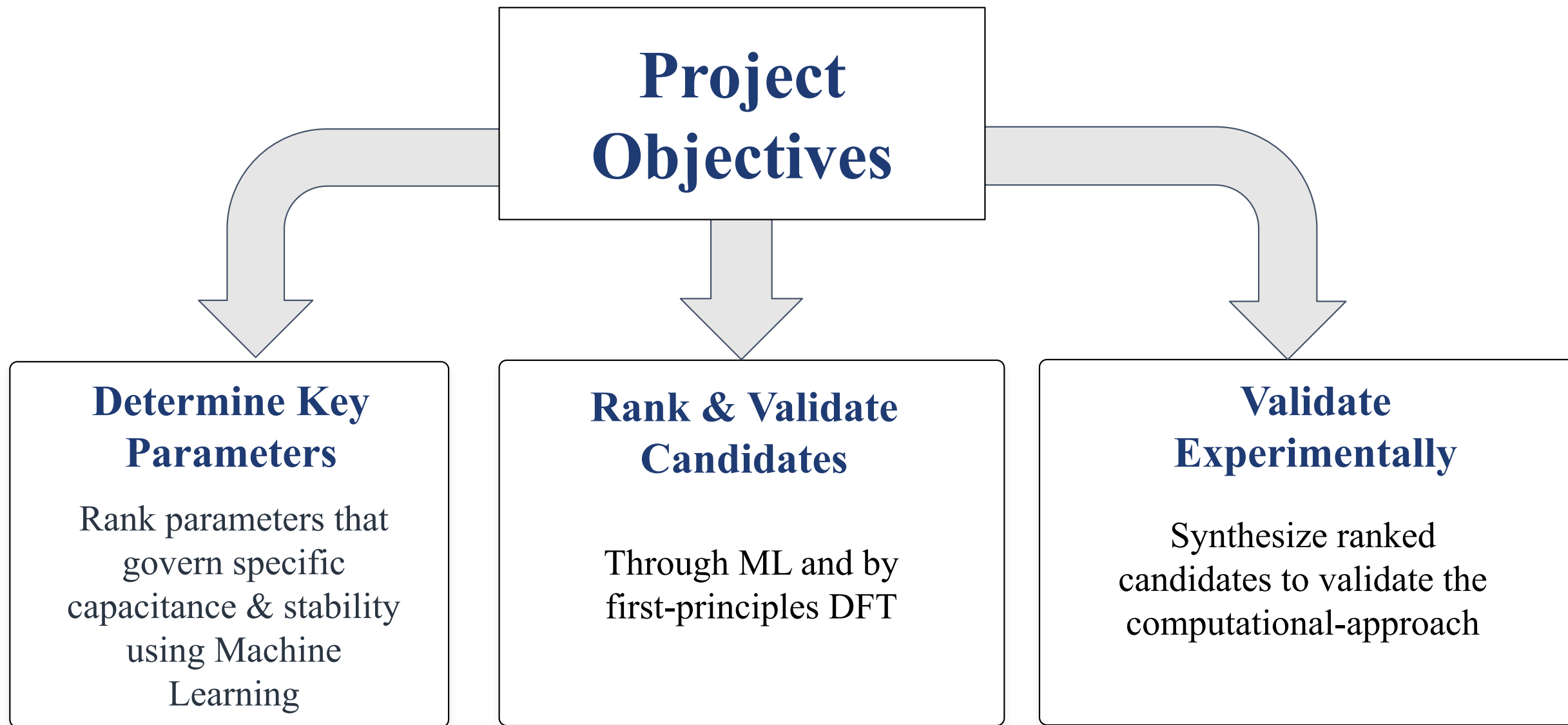
Low Predictive Performance

Research Gap

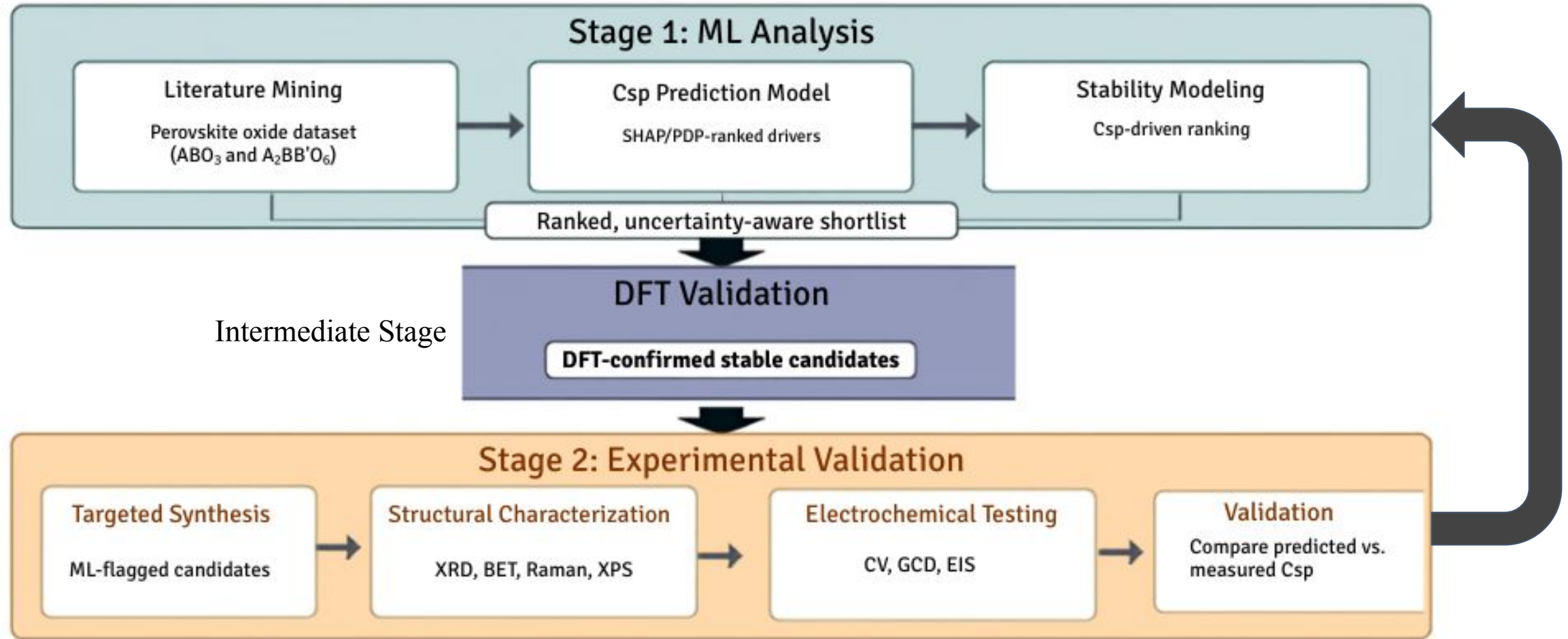
No first-principles validation, No feedback loop

Ghosh, S., Sibi, A., Thomas, T. Machine Learning and Explainable Artificial Intelligence Reveal Physical Insights into Biomass-Derived Carbon for High-Performance Supercapacitors. *J. Power Sources* **2026**, 665, 239040. DOI: 10.1016/j.jpowsour.2025.239040.

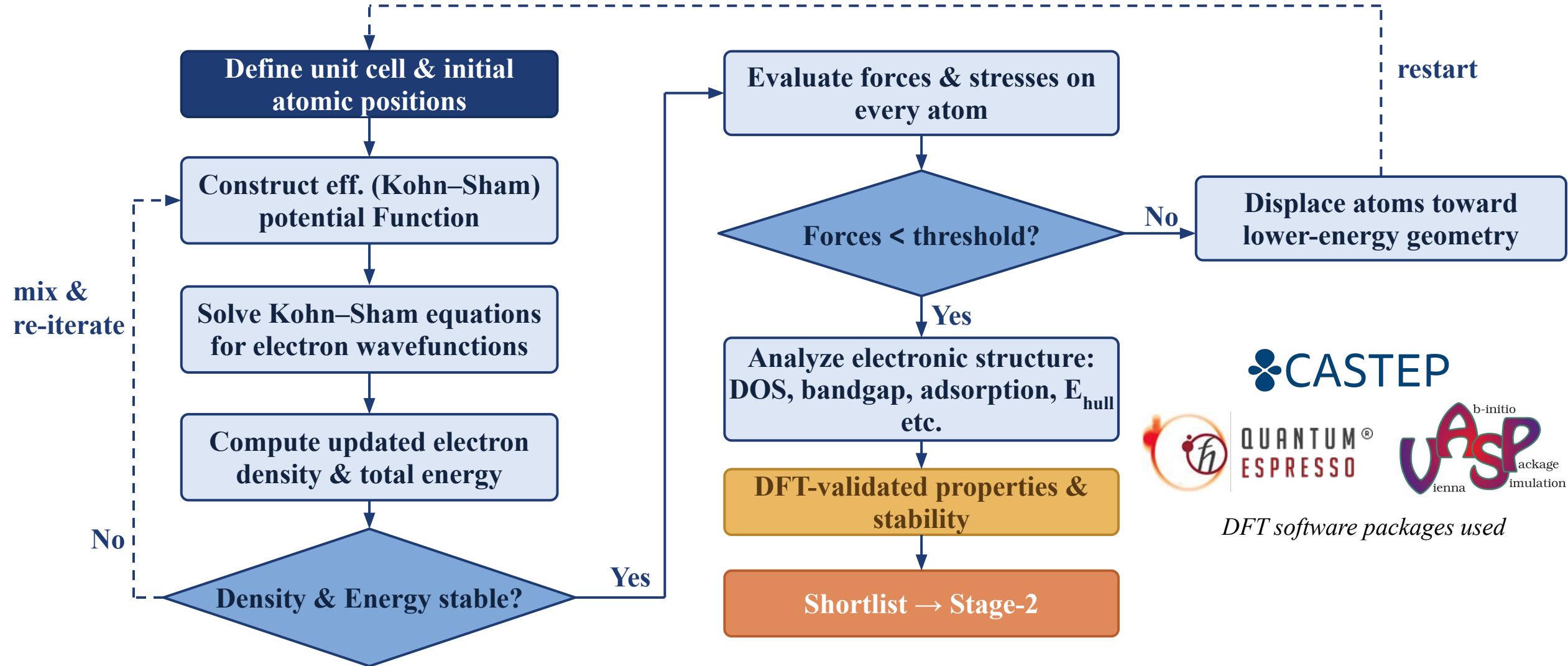
Ghosh, S., Rao, G. R. & Thomas, T. Machine Learning-Based Prediction of Supercapacitor Performance for a Novel Electrode: Cerium Oxynitride. *Energy Storage Materials* **2021**



Proposed Solution: Computation-guided Closed-Loop Pipeline



Density Functional Theory (DFT) Workflow



Density Functional Theory (DFT) Modeling

Property	How It Is Estimated
Band structure / bandgap	Kohn–Sham eigenvalues along k-path
Density of States (DOS)	Electron occupation vs. energy near Fermi level
Formation / cohesive energy	Total energy difference vs. reference states
Adsorption / binding energy	Energy difference before/after adsorbate binding
Work function	Vacuum potential minus Fermi energy (slab calculation)
Charge density distribution	Bader charge analysis on converged electron density
Lattice parameters / elastic constants	Geometry and energy optimization (structural relaxation)
Phonon dispersion / stability	Density Functional Perturbation Theory (DFPT)

Measurement Techniques

Experiment / Process	Property Determined
XRD (X-ray Diffraction)	Crystal structure and phase identification
Raman Spectroscopy	Vibrational modes, bonding, and structural defects
FTIR	Functional groups and chemical bonding
SEM	Surface morphology
FESEM	High-resolution surface morphology and nanostructure
TEM	Internal structure, particle size, and lattice fringes
XPS	Surface elemental composition and oxidation states
CV (Cyclic Voltammetry)	Redox behavior and specific capacitance
GCD (Galvanostatic Charge-Discharge)	Specific capacitance, cycling stability, coulombic efficiency
EIS (Electrochemical Impedance Spectroscopy)	Charge transfer resistance and ion diffusion
BET	Surface area and pore size distribution
UV-Vis Spectroscopy	Optical absorption and bandgap
Zeta Potential	Surface charge and colloidal stability

Measurement Methods: Key Electrode Properties

Property	Technique	How It Is Measured
Specific Capacitance	CV (Cyclic Voltammetry)	
Specific Capacitance	GCD (Galvanostatic Charge-Discharge)	
Surface Area	BET	N ₂ adsorption-desorption isotherm
Pore Volume	BJH / Single-point adsorption	Adsorbed volume at $P/P_0 \approx 0.99$
Pore Size Distribution	BJH / DFT model	Applied to the desorption branch
Porosity	Gas adsorption + density	From pore volume and skeletal/bulk density

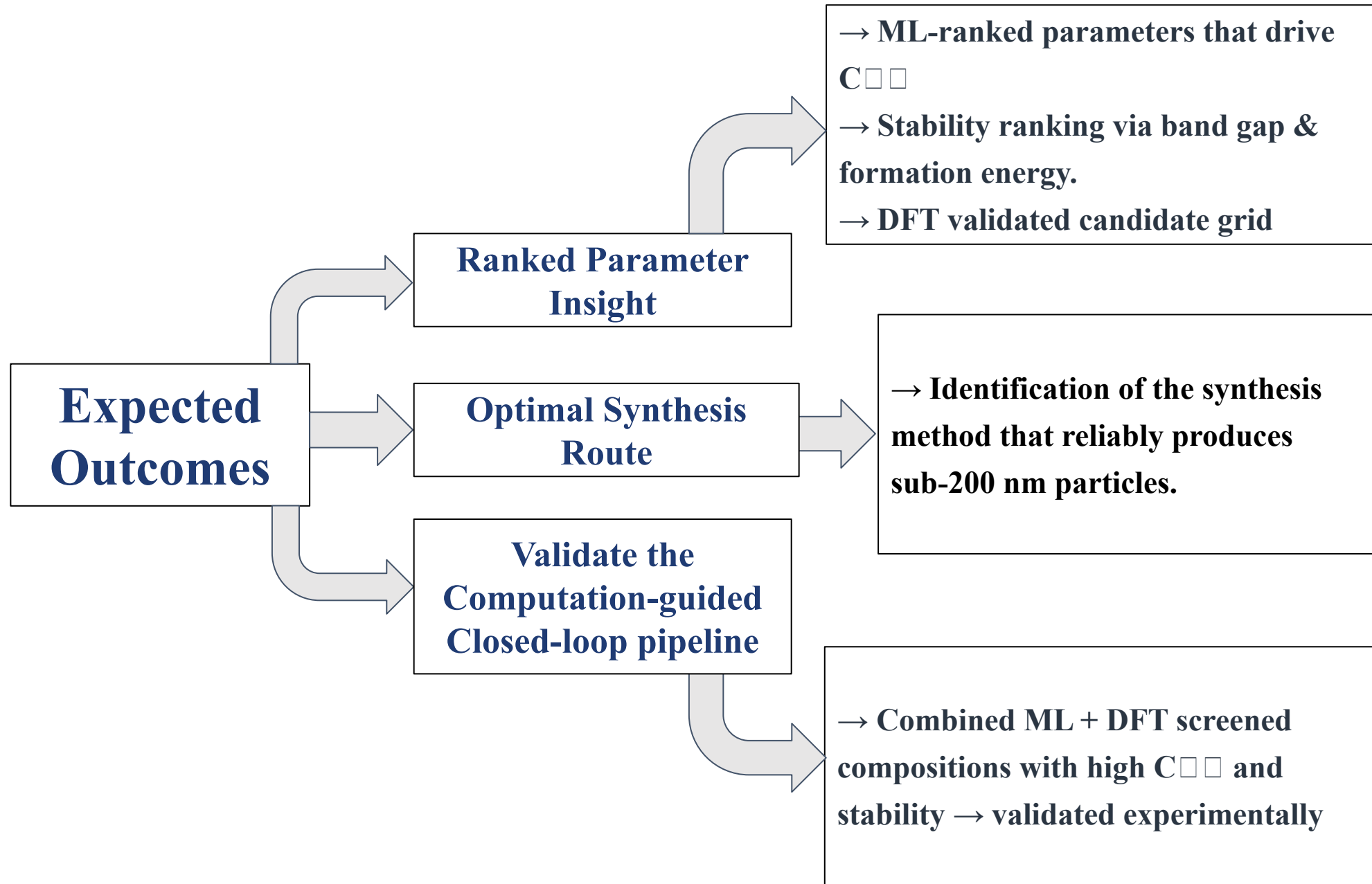
Two-stage Project Execution Plan

Stage-1: Computational Analysis

- **ML-guided ranking**
- **First-principles DFT validation**
- **Iterate and feedback ML ↔ DFT sequence**
until both agree on ranked candidates and synthesizability

Stage-2: Experimental Execution

- **Nanoscale synthesis:** find out suitable processes to target *sub-200 nm particles* (e.g. sol-gel, combustion, hydro/solvothermal etc.)
- **Synthesis & characterization**
- **Computational predictions validated against experiment; and closing the loop back to stage-1.**



Work Done in Semester-1 | Overview

- **Built a usable dataset** — literature-mined ABO_3 perovskite 257 rows (189 GCD across 29 papers)
- **Prepared a classification-based ML model ensemble** ranking each composition's propensity to reach top-tier C_{sp} (top-33%, > 339.7 F/g). [The model = equal weight ensemble of CatBoost + HistGradientBoosting + XGBoost]
- **Applied explainability (SHAP/PDP)** — identified the key drivers of high C_{sp}
- **Produced a ML ranked shortlist** — both undoped and doped candidate compositions

Next step: DFT-based validation → shortlist candidates for stage-2 → synthesis and characterization → Feedback

Work Done in Semester-1 | The C_{sp} -predictive Dataset

→ Prepared a usable, literature-mined dataset of 257 ABO_3 perovskite compositions (189 GCD-measurement rows across 29 papers), to rank candidates based on C_{sp}

Feature space — organized into four descriptor families:

Chemistry

mean melting-T, mean electronegativity, ionic radii (rA/rB), Nd-valence spread

Structure

crystal phase, tolerance factor, octahedral factor, BET surface area

Processing

synthesis method

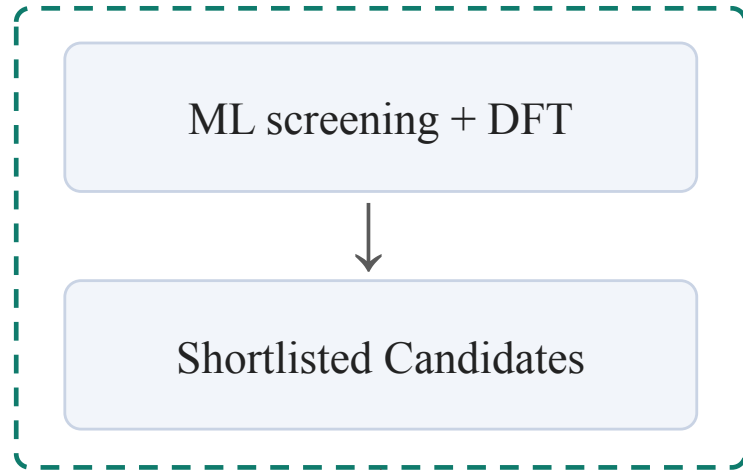
Condition

electrolyte, current density, concentration, scan rate

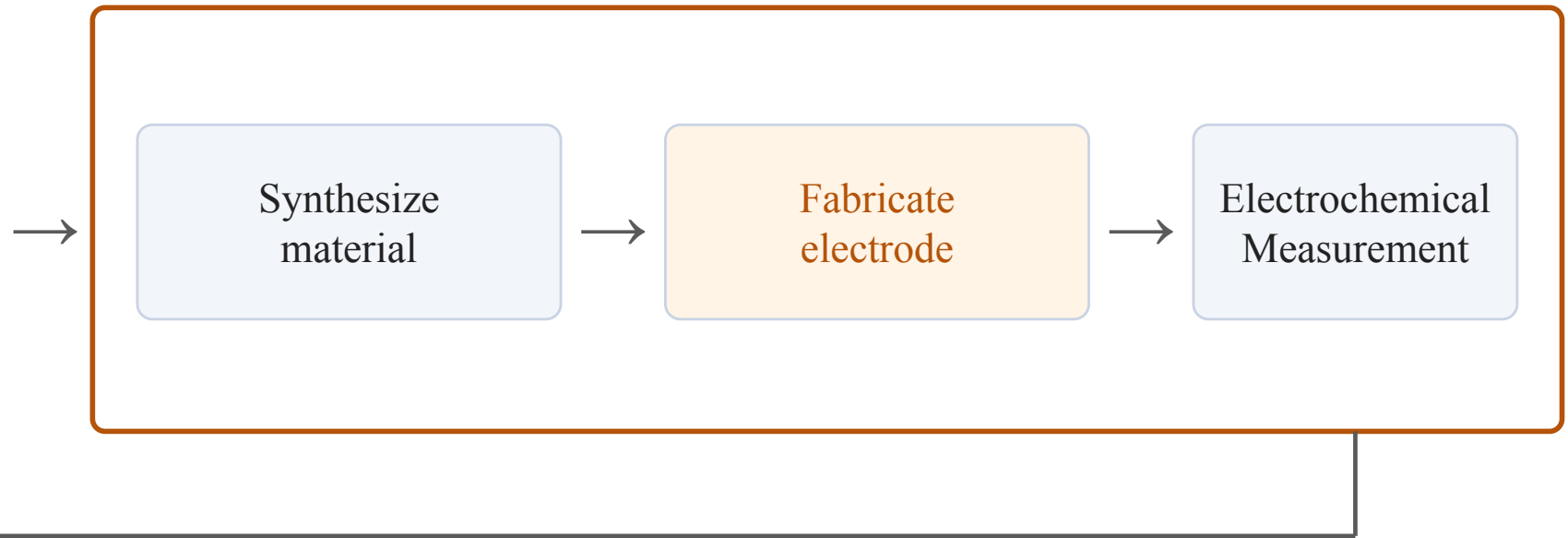
Target variable: gravimetric C_{sp} (unit: F/g)

Electrode Fabrication in Stage-2

Stage-1: Computational Analysis



Stage-2: Experimental validation



Feedback: measurements refine the model

Two **possible** routes to fabricate the electrode:

1. Spray Pyrolysis Technique
2. Laser Induced Graphene (LIG) Technique

Spray Pyrolysis Technique

[Synthesis + fabrication done in one step]

1. Prepare precursor solution
2. Load into spray gun / nozzle
3. FTO* glass on hotplate, $\sim 300\text{--}500\text{ }^{\circ}\text{C}$
4. Spray in short bursts*, fixed distance
5. Precursor decomposes on contact
6. Oxide film grows $\rightarrow$ anneal

*Bursts, not continuous — a steady spray cools the plate.

*FTO = the transparent conducting substrate.



Spray Pyrolysis Machine



Hot Spray Pyrolysis Machine
(STRC)

Laser Induced Graphene (LIG) Technique

1. Mount polyimide film
2. CO₂ laser writes pattern
3. Local temp. > 2500 °C
4. Non-carbon atoms leave as gas
5. Carbon → porous 3D graphene
6. Drop-cast oxide + Nafion → dry



PROF. DR. SAEED
MAHMUD ULLAH

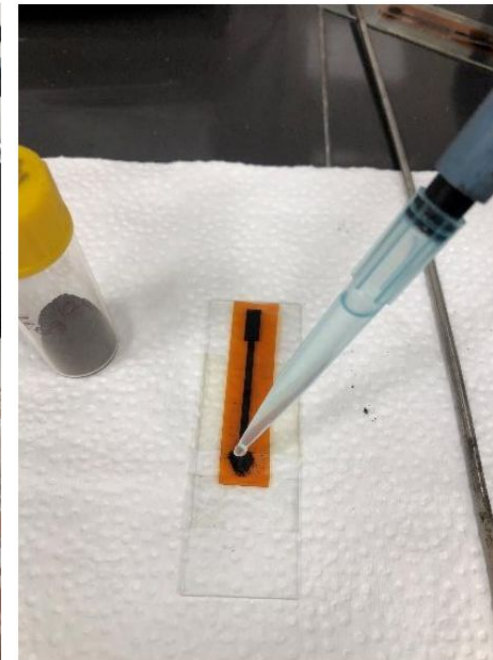


DR. MD. AHSAN
HABIB

Microsystem & Nanoengineering Lab



Laser engraving of
electrode

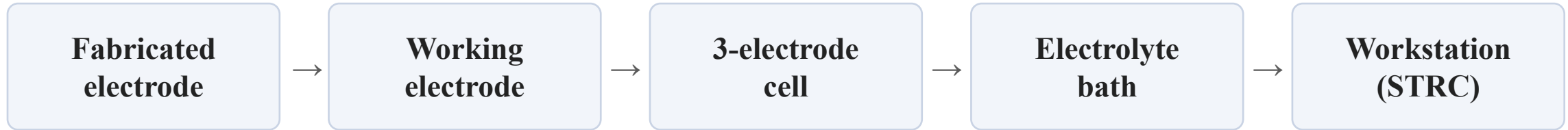


Applying Nafion binder
to improve adhesion

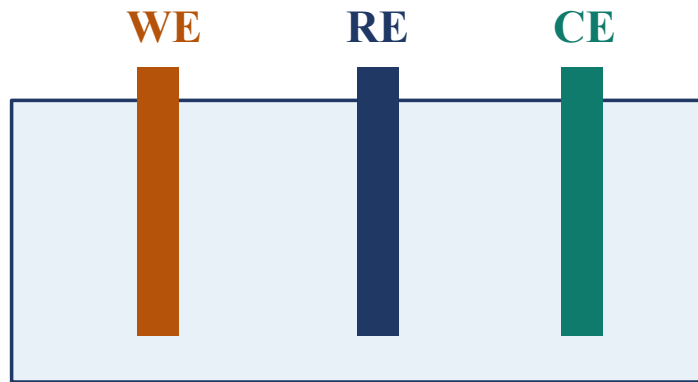


Fabricated electrode
after drying

Electrochemical Measurement Setup



Three-Electrode Cell



WE = working electrode

RE = reference electrode (Ag/AgCl)

CE = counter electrode (Pt)

Cyclic Voltammetry (CV)

Capacitance from curve area

Galvanostatic Charge–Discharge (GCD)

Capacitance from discharge slope

Electrochemical Impedance Spectroscopy

Charge-transfer resistance; ion diffusion

Key Takeaways

- Semester 1 focused **mainly** on planning & few groundworks
- Work done: A descriptor-dataset space and The computation-guided closed-loop pipeline design
- Semester-2 moves to execution: ML/DFT screening → synthesis → characterization → validation

❑ Funding

- i. MSRL Research Assistantship, University Grants Commission (UGC) of Bangladesh
- ii. Ministry of Science and Technology (Bangladesh)

❑ Acknowledgements

- Material Science Research Laboratory (MSRL), Dept. of EEE, University of Dhaka
- Semiconductor Technology Research Centre (STRC), University of Dhaka
- Nanophysics & Soft Matter Lab, Dept. of Physics, University of Dhaka
- Centre for Advanced Research in Sciences (CARS), University of Dhaka
- Atomic and Energy Commission (AEC), BAEC
- Bangladesh Council of Scientific and Industrial Research (BCSIR)



Thank You

Work Done in Semester-1 | ML Model

→ **Built a classification-oriented ML model** that ranks compositions by their propensity to reach top-tier specific capacitance (top-33%, $C_{sp} > 339.7$ F/g).

Chosen model ensemble:

Ensemble	ROC-AUC	PR-AUC
Cat + Hist + XGB (equal weighted)*	0.806	0.661
Cat + Hist	0.807	0.652

***Chosen:** 3-way diversity gives the best PR-AUC and most robust ranking. Progression: 0.762 (best single model) → 0.806 (ensemble).

Validated with paper-grouped 5-fold CV (no source-paper leakage); no-skill floor at $ROC = 0.500$.

Work Done in Semester-1 | Explainability (XAI)

→ **Applied SHAP explainability** to the ensemble (out-of-fold) to identify which features drive high-Csp propensity.

Top drivers among several others:

Feature	Family	Effect on high-Csp likelihood
mean melting-T	Chemistry	Higher → more likely (robust M–O bonding / redox lattice)
current density	Condition	Higher → less likely (rate limitation)
synthesis method	Processing	Governs morphology & surface area
mean electronegativity	Chemistry	Bonding character / redox activity
crystal phase	Structure	Structural family of the composition

These drivers reappeared independently in the earlier regression work → the signal is real chemistry, not an artifact.

Work Done in Semester-1 | DFT Hand-off Candidates

→ **Top-ranked chemistry-in-domain candidates handed to DFT validation** (scored at fixed conditions: 1 A/g, KOH, GCD, 1 M).

Top 5 Undoped Candidates

Composition	pctile	$\pm$ SD	risk	DFT
SrCoO ₃	99	0.24	support.	P1
LaCoO ₃	83	0.23	support.	P1
LaCrO ₃	73	0.23	support.	P1
SrMnO ₃	62	0.23	support.	P1
GdNiO ₃	54	0.21	support.	P1

All P1 (data-supported); genuine score gradient (0.39 → 0.26).

Top 5 Doped Candidates

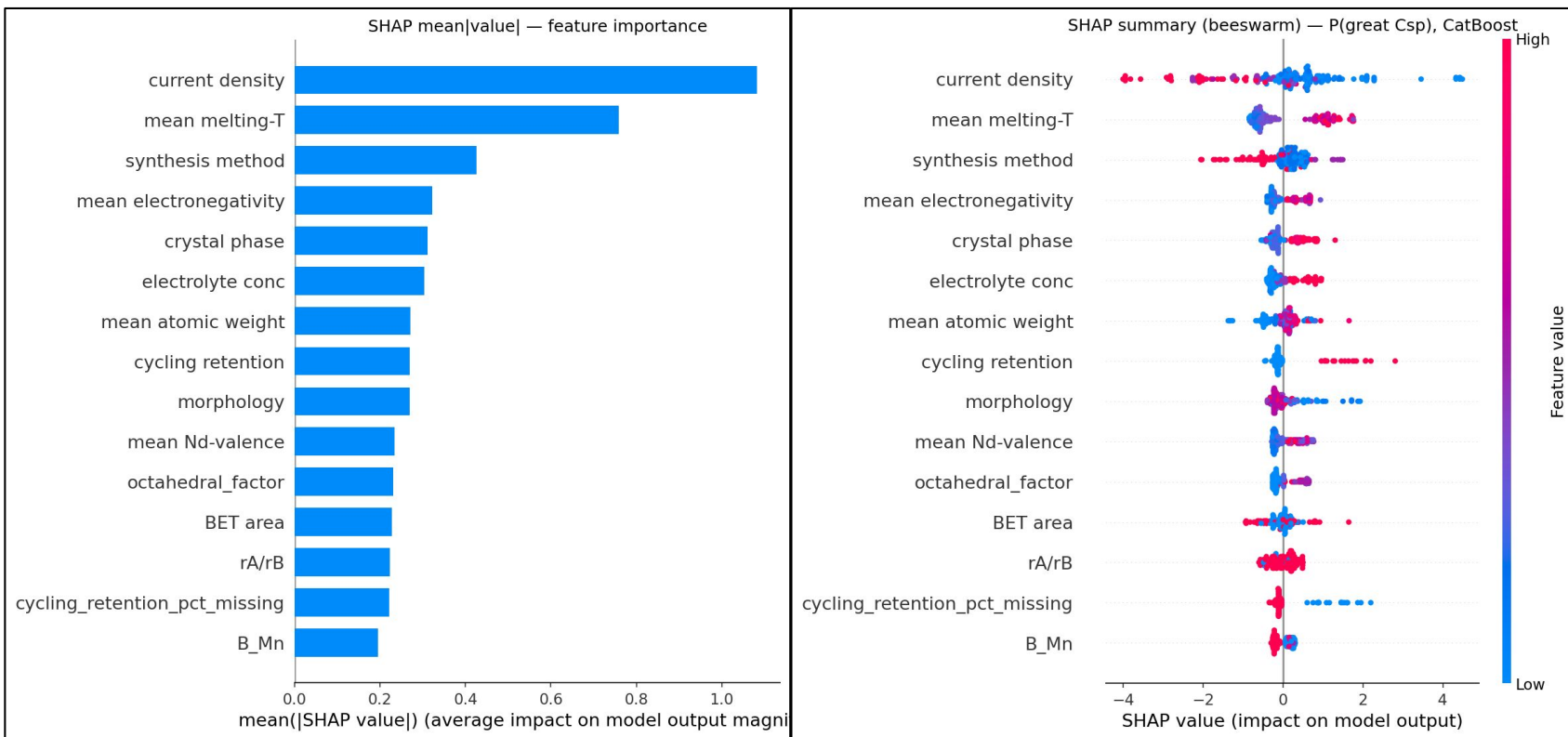
Composition	x	pctile	$\pm$ SD	risk	DFT
BiCo _{0.9} Ni _{0.1} O ₃	0.1	100	0.22	extrap.	P2
SrCo _{0.9} Ni _{0.1} O ₃	0.1	100	0.24	support.	P1
BiCr _{0.9} Co _{0.1} O ₃	0.1	100	0.22	extrap.	P2
SrCr _{0.9} Co _{0.1} O ₃	0.1	100	0.24	extrap.	P2
CaCo _{0.9} Ni _{0.1} O ₃	0.1	100	0.23	extrap.	P2

Scores within one $\pm$ SD ($\approx$ 0.22) → statistically tied; select by risk / DFT priority.

P1 = data-supported, validate first **P2** = extrapolated, DFT checks if real • score = ordinal Csp-propensity (not a probability)

Work Done in Semester-1 | Key Findings (SHAP)

→ **SHAP analysis of the ensemble reveals which features drive high-Csp propensity — and in which direction.**



What the curves show

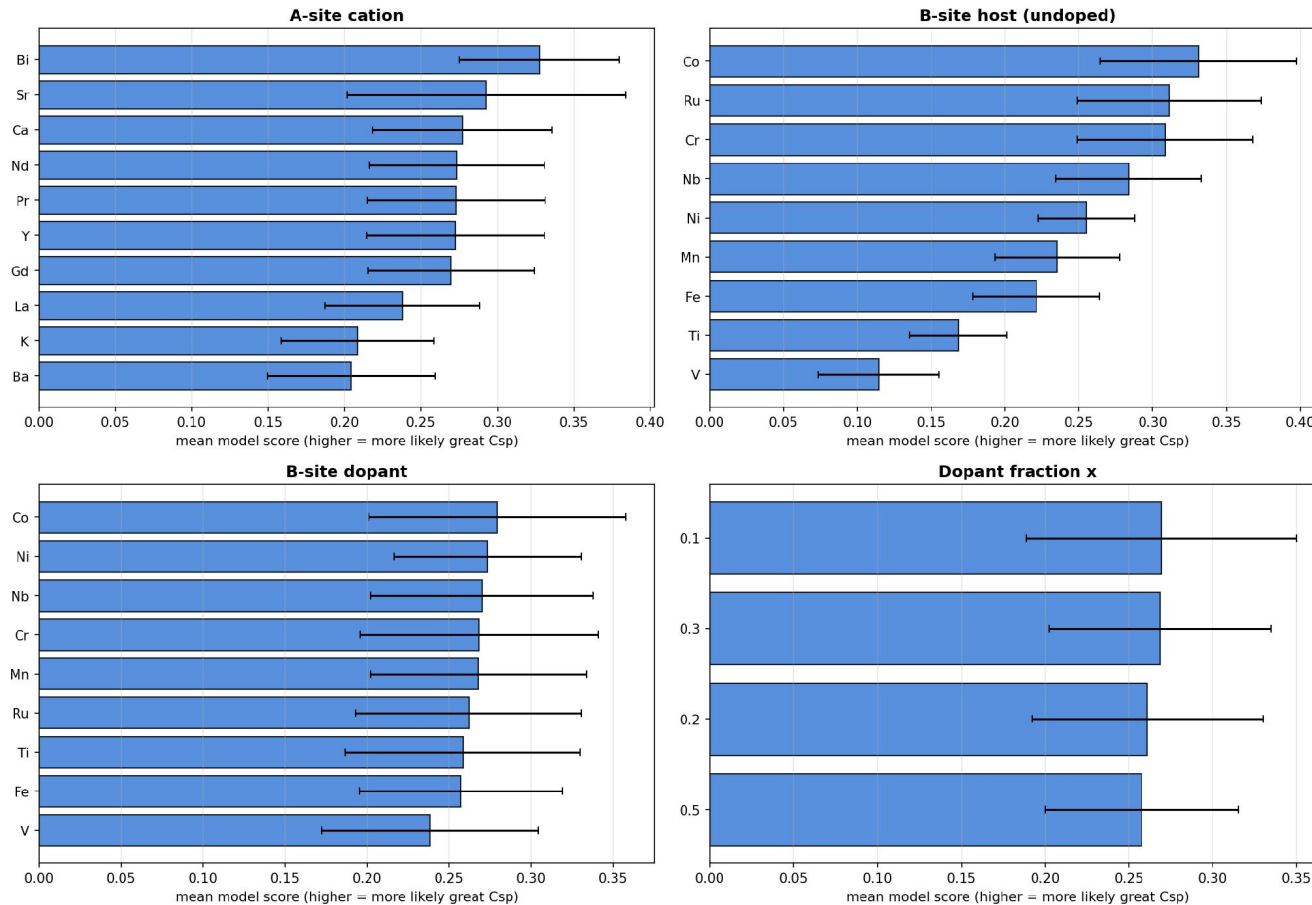
Importance ranking (left): current density and mean melting-T dominate, then synthesis method, electronegativity, crystal phase.

Direction (right, beeswarm): red = high feature value. High melting-T pushes toward great Csp; high current density pushes away.

Work Done in Semester-1 | Key Findings (Rankings & Effects)

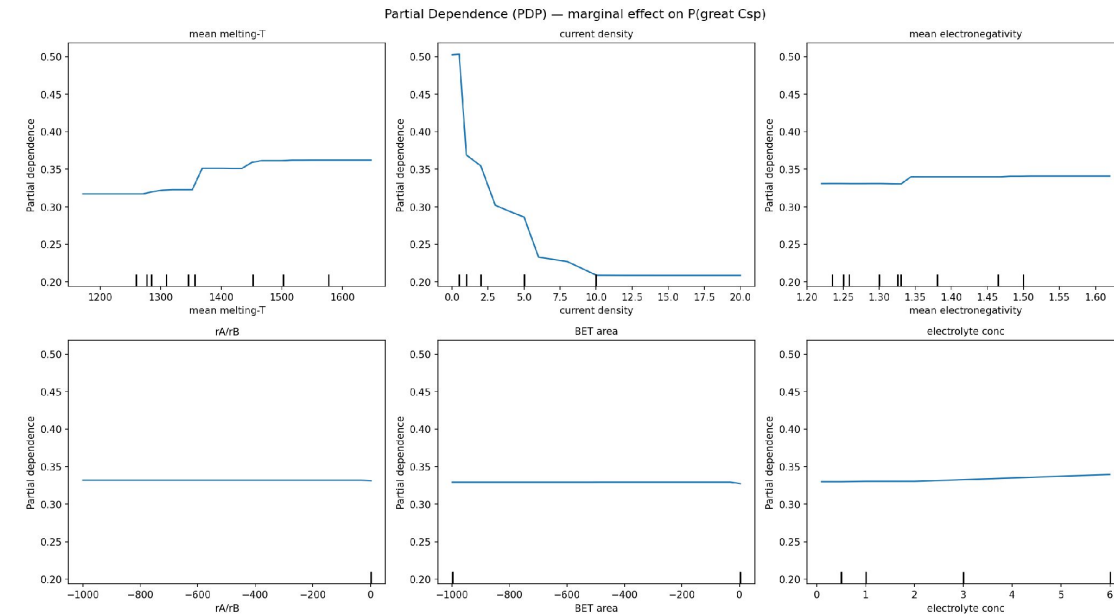
Factor-level rankings — which chemistry levers matter most

Factor-level ranking (marginalized over candidate grid) — screening at 1 A/g, KOH, GCD, 1 M



→ B-site host is the strongest lever (Co > Ru > Cr ..., widest separation); A-site second (Bi > Sr > Ca ...). Dopant identity and fraction x are fine-tuning only. Points to Sr/Bi-cobaltites.

Partial dependence — marginal effect on Probability (of being in great C_{sp} zone)



Confirms SHAP directions: melting-T rises then plateaus; current density falls sharply — the two dominant, physically-correct effects.

Caveat: PDP assumes feature independence and is noisy in sparse regions (flat rA/rB , BET tails) — read as indicative, not causal.

Why Sub-200 nm? Why Bulk Fails

Surface-limited

Pseudocapacitance is a near-surface Faradaic reaction — only the outer ~20–50 nm (ion diffusion length, $\sqrt{Dt}$) actually stores charge.

Dead mass

The inactive core still counts in the denominator of gravimetric C_{grav} (F/g) but stores nothing. Utilisation fraction: ~20% (bulk) $\rightarrow$ ~70% (sub-200 nm).

Rate capability

Shorter ion diffusion path — and since $\tau \propto L^2/D$, halving the particle size cuts the diffusion time by 4 $\times$. Better retention at high current density.

Sub-200 nm raises the surface-area-to-volume ratio ($\propto 1/R$), so far more of the mass is accessible.